A cantilever with three elements, written out in full

MATH 6660 — companion note to Homework 3

Notation, all of it, before we start

Everything in this note is one of these eight symbols. The examples refer to the three-element beam set up in the next section.

symbol	say it out loud as	what kind of thing	example here
h	"element length"	one number	$h = \frac{1}{3}$
x_0, x_1, x_2, x_3	"node positions"	four numbers	$0,\ \frac{1}{3},\ \frac{2}{3},\ 1$
ξ_2	"**deflection** unknown at node 2"	a number (unknown)	$\xi_2 = U(2/3)$
η_2	"**slope** unknown at node 2"	a number (unknown)	$\eta_2 = U'(2/3)$
$\phi_2^{(1)}$	"**deflection shape** of node 2"	a function of x	bump at 2/3, flat top
$\phi_2^{(2)}$	"**slope shape** of node 2"	a function of x	crosses zero at 2/3
U	"the FEM answer"	a function of x	the six terms in §3
u	"the exact answer"	a function of x	$x^2(x^2 - 4x + 6)/24$

The two decorations on ϕ are the only thing to memorise:

$\phi_{\textbf{node}}^{(\textbf{kind})}$ **subscript** = which node (1, 2 or 3).
superscript in brackets = which kind: (1) deflection shape, (2) slope shape.
The superscript is not a power and not a derivative.

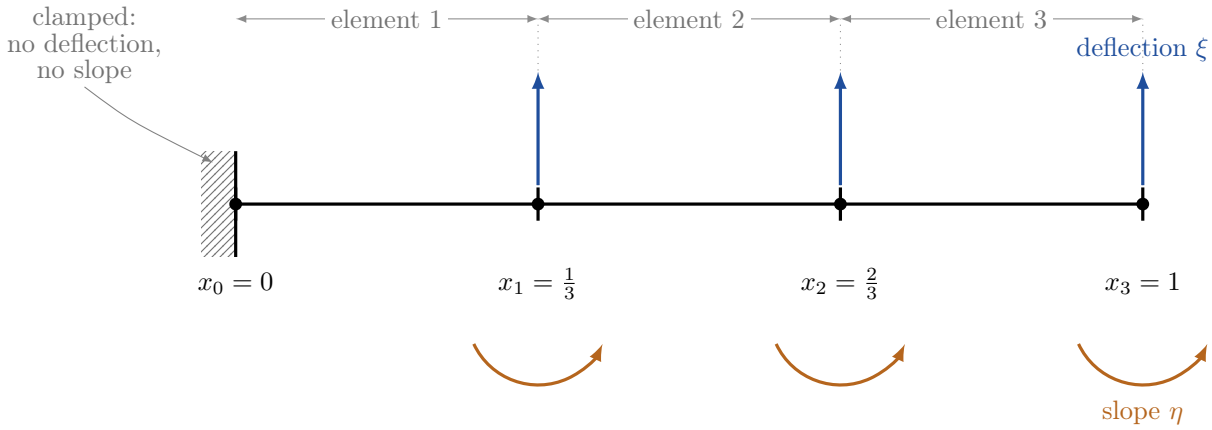
Greek letters: ξ ("ksi") and η ("eta") are just two different letters, chosen so you can tell a deflection unknown from a slope unknown at a glance. They carry *one* subscript: the node number.

On the letter L. An earlier draft of the solutions wrote things like $L_2^{(1)}(v)$. That was my own shorthand, it is *not* used in the course, and it has been removed. If you meet it: $L_2^{(1)}(v)$ means "measure the value of v at node 2" and $L_2^{(2)}(v)$ means "measure the slope of v at node 2". Nothing more.

0. The mesh

Cantilever beam, clamped at the left end, free at the right end, $u^{(iv)} = f$ on $(0, 1)$. Chop it into **three equal elements**, so

$$h = \tfrac{1}{3}, \qquad x_0 = 0, \quad x_1 = \tfrac{1}{3}, \quad x_2 = \tfrac{2}{3}, \quad x_3 = 1.$$



Which node keeps its unknowns.

node	position	what the problem says	unknowns there
x_0	0	clamped: $u(0) = 0$ and $u'(0) = 0$	**none** — both are known to be 0
x_1	1/3	nothing	deflection, slope
x_2	2/3	nothing	deflection, slope
x_3	1	free end: no deflection or slope is prescribed	deflection, slope

So there are **six unknown numbers**. Everything below is a list of six things.

1. The six unknowns

Two names, because there are two *kinds* of unknown:

$\underbrace{\xi_1,\ \xi_2,\ \xi_3}$
deflections (how far the beam moved)

$\underbrace{\eta_1,\ \eta_2,\ \eta_3}$
slopes (how much it tilted)

symbol	read it as	it equals
ξ_1	"deflection at node 1"	$U(1/3)$
ξ_2	"deflection at node 2"	$U(2/3)$
ξ_3	"deflection at node 3"	$U(1)$
η_1	"slope at node 1"	$U'(1/3)$
η_2	"slope at node 2"	$U'(2/3)$
η_3	"slope at node 3"	$U'(1)$

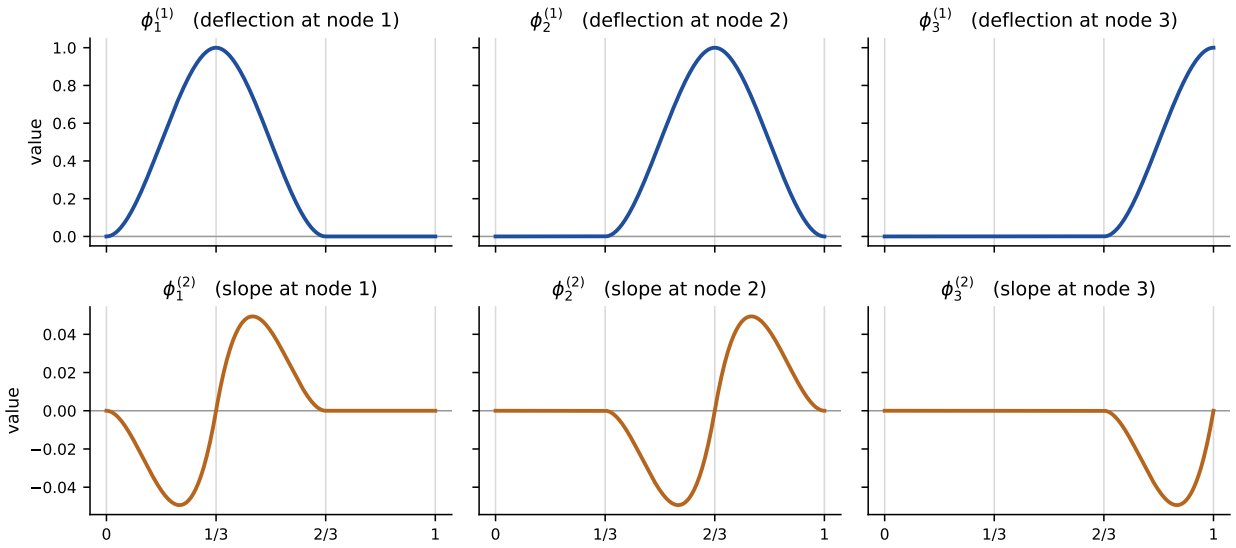
The subscript is **the node number and nothing else**. ξ and η are just two different letters so you can tell a deflection from a slope.

2. The six basis functions

Each unknown needs one function to carry it, so there are six of those too — $\phi_{\text{node}}^{(\text{kind})}$, as set out on page 1. The complete list is

$$\underbrace{\phi_1^{(1)},\ \phi_2^{(1)},\ \phi_3^{(1)}}_{\text{deflection shapes}} \qquad \underbrace{\phi_1^{(2)},\ \phi_2^{(2)},\ \phi_3^{(2)}}_{\text{slope shapes}}$$

and here they are, drawn on this exact mesh:



Read the pictures:

- $\phi_2^{(1)}$ (top middle) is a bump of height 1 at $x = 2/3$, with a *flat top*, dying at $x = 1/3$ and $x = 1$. It moves node 2 up by one unit and does nothing else.
- $\phi_2^{(2)}$ (bottom middle) passes through *zero* at $x = 2/3$ but crosses with slope 1. It tilts node 2 without moving it.
- $\phi_3^{(1)}$ and $\phi_3^{(2)}$ (right column) are **half** the width of the others: node 3 is the free tip, so it has an element on its left only.

3. The approximate solution, written out

No summation sign — there are only six terms:

$$\boxed{U(x) = \xi_1\phi_1^{(1)}(x) + \xi_2\phi_2^{(1)}(x) + \xi_3\phi_3^{(1)}(x) + \eta_1\phi_1^{(2)}(x) + \eta_2\phi_2^{(2)}(x) + \eta_3\phi_3^{(2)}(x)}$$

4. Why the shapes are built that way: the table is an identity matrix

Take each of the six functions and make all six measurements on it — its value at the three nodes and its slope at the three nodes. (Computed, not asserted.)

	value at $\frac{1}{3}$	value at $\frac{2}{3}$	value at 1	slope at $\frac{1}{3}$	slope at $\frac{2}{3}$	slope at 1
$\phi_1^{(1)}$	**1**	0	0	0	0	0
$\phi_2^{(1)}$	0	**1**	0	0	0	0
$\phi_3^{(1)}$	0	0	**1**	0	0	0
$\phi_1^{(2)}$	0	0	0	**1**	0	0
$\phi_2^{(2)}$	0	0	0	0	**1**	0
$\phi_3^{(2)}$	0	0	0	0	0	**1**

It is the 6×6 identity matrix. That is the whole design of the basis, and it is the only reason the unknowns mean anything. Put $x = 2/3$ into U and read down the "value at 2/3" column: five of the six numbers are zero, one is 1, so

$$U(2/3) = \xi_1\cdot 0 + \xi_2\cdot \mathbf{1} + \xi_3\cdot 0 + \eta_1\cdot 0 + \eta_2\cdot 0 + \eta_3\cdot 0 = \xi_2.$$

Differentiate and read down the "slope at 2/3" column the same way: $U'(2/3) = \eta_2$.

If the table were *not* the identity — say $\phi_2^{(2)}$ had value 0.3 at $x = 2/3$ instead of 0 — you would get $U(2/3) = \xi_2 + 0.3\,\eta_2$, and ξ_2 would no longer be "the deflection at node 2". The zeros are what keep the two kinds of unknown from contaminating each other.

5. The matrix

Six unknowns means a 6×6 system. Order them deflections-first: $(\xi_1, \xi_2, \xi_3, \eta_1, \eta_2, \eta_3)$. With $h = 1/3$ the three blocks are numbers, not symbols:

$$\boldsymbol{A} = \begin{bmatrix} 648 & -324 & 0 \\ -324 & 648 & -324 \\ 0 & -324 & \boxed{324} \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 24 & 6 & 0 \\ 6 & 24 & 6 \\ 0 & 6 & \boxed{12} \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 0 & 54 & 0 \\ -54 & 0 & 54 \\ 0 & -54 & \boxed{-54} \end{bmatrix}$$

$$\begin{bmatrix} \boldsymbol{A} & \boldsymbol{C} \\ \boldsymbol{C}^T & \boldsymbol{B} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \\ \eta_1 \\ \eta_2 \\ \eta_3 \end{bmatrix} = \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ b_3^{(1)} \\ b_1^{(2)} \\ b_2^{(2)} \\ b_3^{(2)} \end{bmatrix}, \qquad \boldsymbol{A}_{\text{entry}} = \int_0^1 \big(\phi^{(1)}\big)''\big(\phi^{(1)}\big)''\,\mathrm{d}x, \text{ etc.}$$

The $\boxed{\text{boxed}}$ corner entries are the free tip. Node 3 has only one element attached instead of two, so it collects half of what an interior node collects: 324 instead of 648, 12 instead of 24, and -54 instead of 0.

6. Worked numbers: uniform load $f = 1$

Solving the 6×6 system with $f \equiv 1$ gives

node	x	FEM deflection	exact $u(x)$	FEM slope	exact $u'(x)$
1	1/3	$\xi_1 = 0.022119$	0.022119	$\eta_1 = 0.117284$	0.117284
2	2/3	$\xi_2 = 0.069959$	0.069959	$\eta_2 = 0.160494$	0.160494
3	1	$\xi_3 = 0.125000$	0.125000	$\eta_3 = 0.166667$	0.166667

with $u(x) = \dfrac{x^2(x^2 - 4x + 6)}{24}$ the exact solution. The nodal values agree to 3×10^{-15} — with only three elements.

And the tip numbers are the two formulas from any statics table, with $w = 1$, $L = 1$, $EI = 1$:

$$\xi_3 = \frac{1}{8} = \frac{wL^4}{8EI} \quad \text{(tip deflection)}, \qquad \eta_3 = \frac{1}{6} = \frac{wL^3}{6EI} \quad \text{(tip rotation)}.$$

All numbers in Sections 4, 5 and 6 were computed, not quoted: the nodal table by evaluating the six functions, the blocks by Gauss quadrature against the element integrals, and Section 6 by solving the 6×6 system. Script: `three_elem_basis.py`.